

Week 7: Looking over Path Finding Algorithm and Storage

Overview:

This week focused on setting up and integrating Firebase Storage and Firestore into the application to manage anchor point metadata, navigation paths, and user interactions in real time. The primary objective was to establish a secure and local-based system that connects the admin mapping module with the user navigation interface.

Progress:

- Configured Firebase Realtime Database and Firestore in the Android project.
- Created structured collections to store anchor coordinates, labels, and relationships.
- Integrated Firebase SDK and verified app connection using test data uploads.
- Developed code modules to store and retrieve location data dynamically.
- Tested Firebase Authentication for user login and data access control.
- Ensured real-time syncing between admin-tagged points and user app view.
- Logged and monitored data operations using Firebase Analytics and console logs.

Summary:

Firebase has been successfully implemented to serve as the core data backbone of the system. Real-time anchor updates, station layout storage, and secure data transactions are now functional, enabling dynamic and location-based services in the app.

Next Week and Beyond:

Focus will now shift to integrating Firebase data with the A* pathfinding logic and AR navigation components.